Analisi con SQL

Tutte le query SQL utilizzate per le analisi.

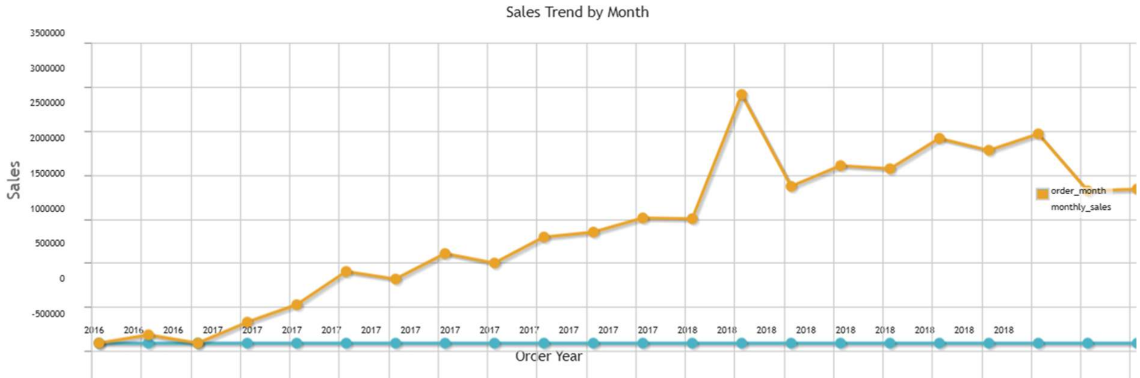
1. Vendite nel tempo

- <u>Andamento del fatturato per mese.</u>

```sql
SELECT
    YEAR(o.order_date) AS order_year,
    MONTH(o.order_date) AS order_month,
    SUM(oi.price) AS monthly_sales
FROM
    orders AS o
JOIN
    order_items AS oi ON o.order_id = oi.order_id
WHERE
    o.order_status IN ('delivered', 'invoiced', 'shipped')
GROUP BY
    YEAR(o.order_date),
    MONTH(o.order_date)
ORDER BY
    order_year,
    order_month;
```

```sql
SELECT YEAR(o.order_date) AS order_year, MONTH(o.order_date) AS order_month, SUM(oi.price) AS monthly_sales FROM orders AS o JOIN order_items AS oi ON o.order_id = oi.order_id WHERE o.order_status IN ('delivered',
'invoiced', 'shipped') GROUP BY YEAR(o.order_date), MONTH(o.order_date) ORDER BY order_year, order_month;
```

order_year 1	order_month 2	monthly_sales
2016	9	895.07
2016	10	94451.06
2016	12	215.94
2017	1	239142.80
2017	2	438050.94
2017	3	815950.08
2017	4	730035.48
2017	5	1020334.81
2017	6	912246.45
2017	7	1208502.45
2017	8	1264232.95
2017	9	1426012.44
2017	10	1416984.31
2017	11	2830664.28
2017	12	1785126.77
2018	1	2021302.46
2018	2	1985676.08
2018	3	2331790.96
2018	4	2196048.06
2018	5	2382707.78
2018	6	1733850.25
2018	7	1754472.42
2018	8	1690655.03
2018	9	155.40



- <u>Identificazione di eventuali pattern di stagionalità o periodi di picco.</u>

```sql
SELECT
    YEAR(o.order_date) AS order_year,
    MONTH(o.order_date) AS order_month,
    SUM(oi.price) AS monthly_sales
FROM
    orders AS o
JOIN
    order_items AS oi ON o.order_id = oi.order_id
WHERE
    o.order_status IN ('delivered', 'invoiced', 'shipped')
GROUP BY
    YEAR(o.order_date),
    MONTH(o.order_date)
ORDER BY
    monthly_sales DESC;
```

```sql
SELECT YEAR(o.order_date) AS order_year, MONTH(o.order_date) AS order_month, SUM(oi.price) AS monthly_sales FROM orders AS o JOIN order_items AS oi ON o.order_id = oi.order_id WHERE o.order_status IN ('delivered',
'invoiced', 'shipped') GROUP BY YEAR(o.order_date), MONTH(o.order_date) ORDER BY monthly_sales DESC;
```

order_year	order_month	monthly_sales 1
2017	11	2830664.28
2018	5	2382707.78
2018	3	2331790.96
2018	4	2196048.06
2018	1	2021302.46
2018	2	1985676.08
2017	12	1785126.77
2018	7	1754472.42
2018	6	1733850.25
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2017	10	1416984.31
2017	8	1264232.95
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2017	6	912246.45
2017	3	815950.08
2017	4	730035.48
2017	2	438050.94
2017	1	239142.80
2016	10	94451.06
2016	9	895.07
2016	12	215.94
2018	9	155.40

2. Clienti

- Top 10 clienti per fatturato complessivo.

```sql
SELECT
        c.customer_id,
        SUM(oi.price) AS customer_total_sales
    FROM customers AS c
    JOIN orders AS o ON c.customer_id = o.customer_id
    JOIN order_items AS oi ON o.order_id = oi.order_id
    WHERE o.order_status IN ('delivered', 'invoiced', 'shipped')
    GROUP BY c.customer_id
    ORDER BY customer_total_sales DESC
LIMIT 10
```

```sql
SELECT c.customer_id, SUM(oi.price) AS customer_total_sales FROM customers AS c JOIN orders AS o ON c.customer_id = o.customer_id JOIN order_items AS oi ON o.order_id = oi.order_id WHERE o.order_status IN ('delivered', 'invoiced', 'shipped') GROUP BY c.customer_id ORDER BY customer_total_sales DESC LIMIT 10;
```

customer_id	customer_total_sales
lwNS6AdlPkdm	6735.00
ondvZDYSibyo	6499.00
Fhl8lkm1yu6D	4799.00
LXBxeq9Kt6by	4799.00
kRm3xxmQrNwq	4799.00
m84WdTDI0jlP	4799.00
1jvmBfVozjRM	4799.00
2tbvqYSvWXnH	4690.00
nvlwohKq2qSl	4690.00
1VsanYKIWAFs	4590.00

- Contributo percentuale dei top clienti rispetto al totale.

```sql
WITH CustomerSales AS (
    -- Calculate total revenue per customer for specific order statuses
    SELECT
        c.customer_id,
        SUM(oi.price) AS customer_total_sales,
        SUM(SUM(oi.price)) OVER() AS grand_total_sales
    FROM customers AS c
    JOIN orders AS o ON c.customer_id = o.customer_id
    JOIN order_items AS oi ON o.order_id = oi.order_id
    WHERE o.order_status IN ('delivered', 'invoiced', 'shipped')
    GROUP BY c.customer_id
)
SELECT
    customer_id,
    customer_total_sales,
    -- Calculate percentage contribution (Customer Sales / Grand Total * 100)
    ROUND((customer_total_sales * 100.0) / grand_total_sales, 3) AS percentage_contribution
FROM CustomerSales
ORDER BY customer_total_sales DESC
LIMIT 10;
```

```sql
WITH CustomerSales AS ( -- Calculate total revenue per customer for specific order statuses SELECT c.customer_id, SUM(oi.price) AS customer_total_sales, SUM(SUM(oi.price)) OVER() AS grand_total_sales FROM customers AS c JOIN orders AS o ON c.customer_id = o.customer_id JOIN order_items AS oi ON o.order_id = oi.order_id WHERE o.order_status IN ('delivered', 'invoiced', 'shipped') GROUP BY c.customer_id ) SELECT customer_id, customer_total_sales, -- Calculate percentage contribution (Customer Sales / Grand Total * 100) ROUND((customer_total_sales * 100.0) / grand_total_sales, 3) AS percentage_contribution FROM CustomerSales ORDER BY customer_total_sales DESC LIMIT 10;
```

customer_id	customer_total_sales	percentage_contribution
lwNS6AdlPkdm	6735.00	0.022
ondvZDYSibyo	6499.00	0.021
kRm3xxmQrNwq	4799.00	0.016
1jvmBfVozjRM	4799.00	0.016
LXBxeq9Kt6by	4799.00	0.016
Fhl8lkm1yu6D	4799.00	0.016
m84WdTDI0jlP	4799.00	0.016
nvlwohKq2qSl	4690.00	0.015
2tbvqYSvWXnH	4690.00	0.015
4e1HjdhOSLGv	4590.00	0.015

3. Prodotti e categorie

- Prodotti e categorie con maggior fatturato e/o quantità vendute.

```sql
SELECT
    p.product_id AS TOP_products, p.product_category,
    SUM(oi.price) AS total_sold
FROM
    products AS p
JOIN
    order_items AS oi ON p.product_id = oi.product_id
GROUP BY
    p.product_id
ORDER BY
    total_sold DESC
LIMIT 20;
```

```sql
SELECT p.product_id AS TOP_products, p.product_category, SUM(oi.price) AS total_sold FROM products AS p JOIN order_items AS oi ON p.product_id = oi.product_id GROUP BY p.product_id ORDER BY total_sold DESC LIMIT 20;
```

TOP_products	product_category	total_sold
9NwzO0Pm0fDM	toys	797298.94
SLTlrWtcYt1m	toys	567053.43
Biwi1BNtUB7l	toys	486503.65
ro08JPncYzLh	garden_tools	467133.53
ZWyg4uNWPHjJ	toys	318890.00
tPlw2QcQOvKf	garden_tools	153366.49
utVPLgd1LM3F	toys	146223.00
GvBzGCvvIC2D	toys	127736.00
5J7az1rwth4i	toys	117385.08
CY8OZ3lT9uqB	telephony	116979.98
7grztYpCkySJ	toys	111454.00
CpeCg2UGCL6P	toys	103836.00
k1G4RHt3R7Nk	toys	100674.00
yhM1TNmnd3og	watches_gifts	98566.00
kktXNGlh1TrV	toys	89588.80
rHSTHjYFnJS8	toys	84733.42
hrjNaMt3Wyo5	toys	83808.84
bfkYf8WcDtMl	toys	82825.05
M9Rq7Nx8etjk	toys	70961.80
LeGxEhH8juyh	watches_gifts	69668.40

```sql
SELECT
    p.product_id AS 'products',
    SUM(oi.price) AS total_sales,
    COUNT(oi.order_id) AS quantity_sold,
    -- Percentage of grand total sales
    ROUND(SUM(oi.price) * 100.0 / SUM(SUM(oi.price)) OVER(), 2) AS sales_percentage,
    -- Percentage of grand total quantity sold
    ROUND(COUNT(oi.order_id) * 100.0 / SUM(COUNT(oi.order_id)) OVER(), 2) AS quantity_percentage
FROM products AS p
JOIN order_items AS oi ON p.product_id = oi.product_id
JOIN orders AS o ON oi.order_id = o.order_id
WHERE o.order_status IN ('delivered', 'invoiced', 'shipped')
GROUP BY p.product_id
ORDER BY total_sales DESC
LIMIT 20;
```

```
SELECT p.product_id AS 'products', SUM(oi.price) AS total_sales, COUNT(oi.order_id) AS quantity_sold, -- Percentage of grand total sales ROUND(SUM(oi.price) * 100.0 / SUM(SUM(oi.price)) OVER(), 2) AS
sales_percentage, -- Percentage of grand total quantity sold ROUND(COUNT(oi.order_id) * 100.0 / SUM(COUNT(oi.order_id)) OVER(), 2) AS quantity_percentage FROM products AS p JOIN order_items AS oi ON p.product_id =
oi.product_id JOIN orders AS o ON oi.order_id = o.order_id WHERE o.order_status IN ('delivered', 'invoiced', 'shipped') GROUP BY p.product_id ORDER BY total_sales DESC LIMIT 20;
```

products	total_sales 1	quantity_sold	sales_percentage	quantity_percentage
9NwzO0Pm0fDM	797298.94	383	2.64	0.43
SLTlrWtcYt1m	566543.44	320	1.88	0.36
Biwi1BNtUB7l	486503.65	295	1.61	0.33
ro08JPncYzLh	467133.53	290	1.55	0.33
ZWyg4uNWPHjJ	318890.00	110	1.06	0.12
tPlw2QcQOvKf	153366.49	91	0.51	0.10
utVPLgd1LM3F	146223.00	77	0.48	0.09
GvBzGCvvlC2D	126959.00	175	0.42	0.20
5J7az1rwth4i	117385.08	110	0.39	0.12
CY8OZ3lT9uqB	116979.98	42	0.39	0.05
7grztYpCkySJ	111454.00	134	0.37	0.15
CpeCg2UGCL6P	103836.00	64	0.34	0.07
k1G4RHt3R7Nk	100674.00	90	0.33	0.10
yhM1TNmnd3og	98566.00	34	0.33	0.04
kktXNGlh1TrV	89588.80	32	0.30	0.04
rHSTHjYFnJS8	84733.42	117	0.28	0.13
hrjNaMt3Wyo5	83808.84	95	0.28	0.11
bfkYf8WcDtMl	82825.05	101	0.27	0.11
M9Rq7Nx8etjk	70961.80	135	0.24	0.15
LeGxEhH8juyh	69668.40	26	0.23	0.03

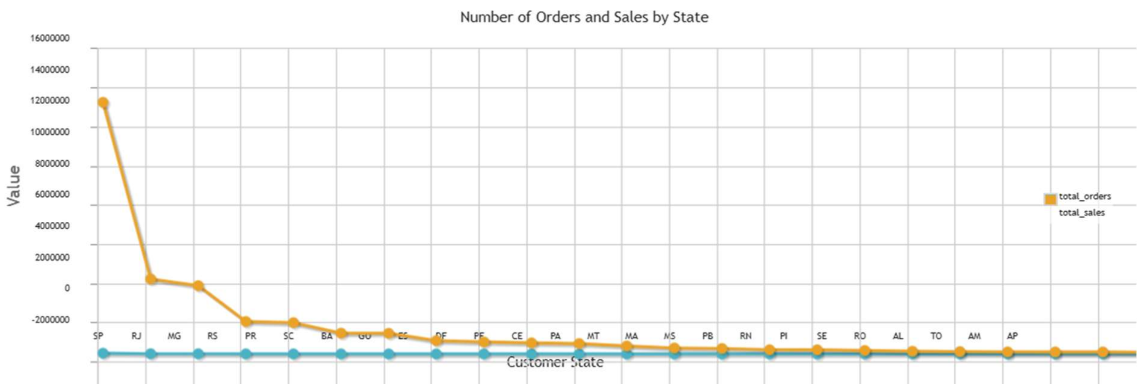
4. Geografia e logistica

- Distribuzione del numero di ordini e del fatturato per lo stato.

```sql
SELECT
    c.customer_state,
    COUNT(DISTINCT o.order_id) AS total_orders,
    ROUND(SUM(oi.price), 2) AS total_sales
FROM customers AS c
JOIN orders AS o ON c.customer_id = o.customer_id
JOIN order_items AS oi ON o.order_id = oi.order_id
WHERE o.order_status IN ('delivered', 'invoiced', 'shipped')
GROUP BY c.customer_state
ORDER BY total_sales DESC;
```

```
SELECT c.customer_state, COUNT(DISTINCT o.order_id) AS total_orders, ROUND(SUM(oi.price), 2) AS total_sales FROM customers AS c JOIN orders AS o ON c.customer_id = o.customer_id JOIN order_items AS oi ON o.order_id
= oi.order_id WHERE o.order_status IN ('delivered', 'invoiced', 'shipped') GROUP BY c.customer_state ORDER BY total_sales DESC;
```

customer_state	total_orders	total_sales 1
SP	37529	12857940.35
RJ	11498	3829469.92
MG	10267	3490441.56
RS	4891	1655613.27
PR	4493	1596990.14
SC	3193	1062730.98
BA	3068	1055729.16
GO	1852	679190.78
ES	1789	613488.88
DF	1780	564504.05
PE	1482	531064.68
CE	1133	408768.03
PA	814	304259.00
MT	846	280171.77
MA	619	214642.08
MS	635	211677.56
PB	493	176829.55
RN	380	137292.37
PI	418	120613.92
SE	305	105508.40
RO	238	101782.91
AL	337	101601.46
TO	246	83978.66
AM	140	45710.75
AP	67	21379.63



- <u>Analisi dei tempi di consegna e del ritardo medio rispetto alla data stimata.</u>

```sql
SELECT
    CASE
        WHEN delivery_date > estimated_delivery_date THEN 'Late'
        WHEN delivery_date <= estimated_delivery_date THEN 'On Time or Early'
        ELSE 'Status Unknown (Missing Date)'
    END AS delivery_status,
    COUNT(*) AS order_count
FROM
    orders
GROUP BY
    delivery_status;
```

```sql
SELECT CASE WHEN delivery_date > estimated_delivery_date THEN 'Late' WHEN delivery_date <= estimated_delivery_date THEN 'On Time or Early' ELSE 'Status Unknown (Missing Date)' END AS delivery_status, COUNT(*) AS
order_count FROM orders GROUP BY delivery_status;
```

delivery_status	order_count
Late	6737
On Time or Early	82170

```sql
SELECT
    AVG(DATEDIFF(delivery_date, estimated_delivery_date)) AS average_delay_for_late_orders
FROM
    orders
WHERE
    delivery_date > estimated_delivery_date;
```

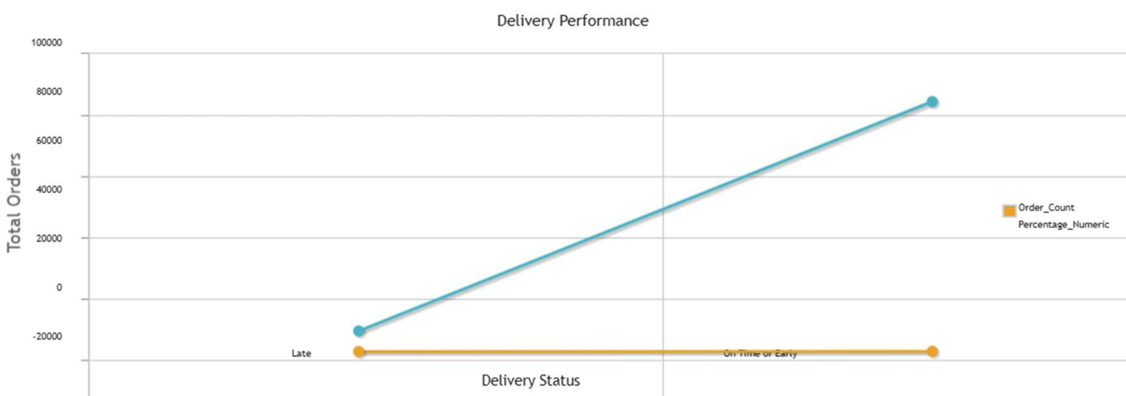
```sql
SELECT AVG(DATEDIFF(delivery_date, estimated_delivery_date)) AS average_delay_for_late_orders FROM orders WHERE delivery_date > estimated_delivery_date;
```

average_delay_for_late_orders
8.6950

```sql
WITH StatusCategorized AS (
    SELECT
        CASE
            WHEN delivery_date > estimated_delivery_date THEN 'Late'
            WHEN delivery_date <= estimated_delivery_date THEN 'On Time or Early'
            ELSE 'Status Unknown (Missing Date)'
        END AS Delivery_Status
    FROM orders
    WHERE order_status IN ('delivered', 'shipped', 'invoice')
)
SELECT
    Delivery_Status,
    COUNT(*) AS Order_Count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS Percentage_Numeric,
    CONCAT(ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2), '%') AS Percentage
FROM StatusCategorized
GROUP BY Delivery_Status
```

```sql
WITH StatusCategorized AS ( SELECT CASE WHEN delivery_date > estimated_delivery_date THEN 'Late' WHEN delivery_date <= estimated_delivery_date THEN 'On Time or Early' ELSE 'Status Unknown (Missing Date)' END AS
Delivery_Status FROM orders WHERE order_status IN ('delivered', 'shipped', 'invoice') ) SELECT Delivery_Status, COUNT(*) AS Order_Count, ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2) AS Percentage_Numeric,
CONCAT(ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 2), '%') AS Percentage FROM StatusCategorized GROUP BY Delivery_Status;
```

Delivery_Status	Order_Count	Percentage_Numeric	Percentage
Late	6737	7.62	7.62%
On Time or Early	81627	92.38	92.38%



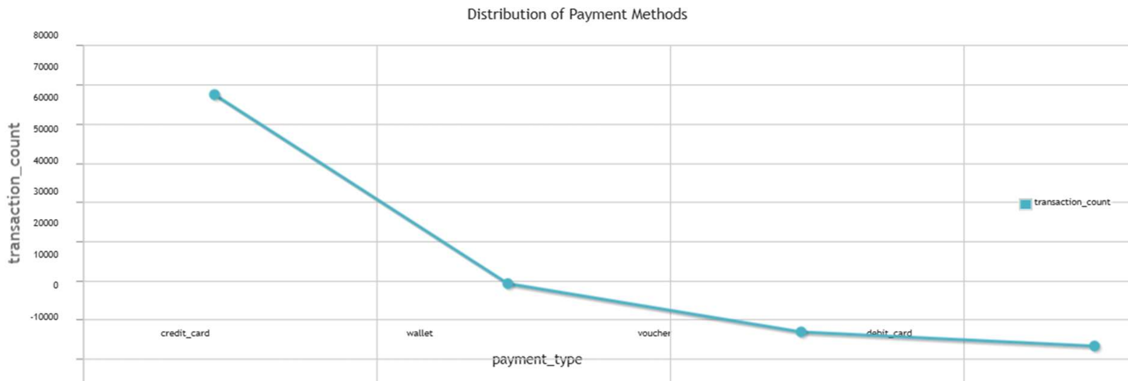
5. Pagamenti

- Distribuzione dei metodi di pagamento.

```sql
SELECT
    payment_type,
    COUNT(*) AS transaction_count
FROM payments
GROUP BY payment_type
ORDER BY transaction_count DESC;
```

```sql
SELECT payment_type, COUNT(*) AS transaction_count FROM payments GROUP BY payment_type ORDER BY transaction_count DESC;
```

payment_type	transaction_count
credit_card	65511
wallet	17227
voucher	4886
debit_card	1283

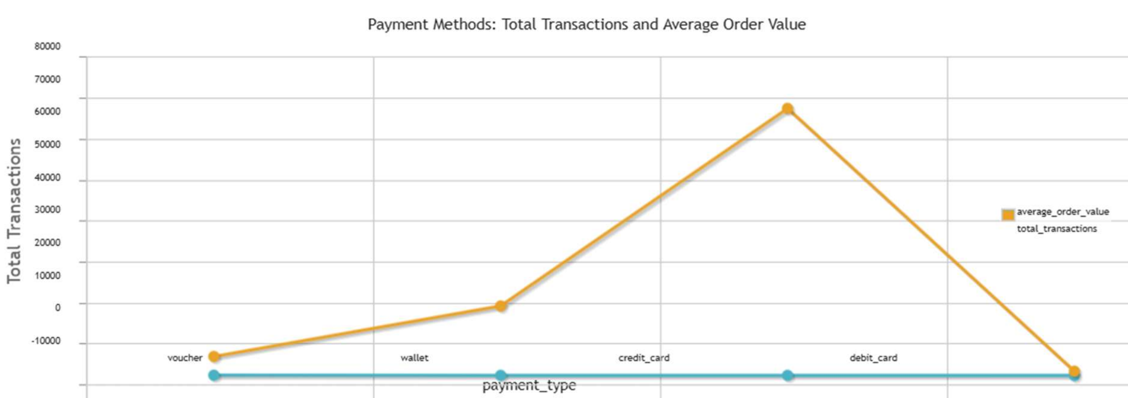


<u>- Relazione tra modalità di pagamento (rate, contanti, carta, ecc.) e valore medio dell’ordine.</u>

```sql
SELECT
    payment_type,
    ROUND(AVG(payment_value), 2) AS average_order_value,
    COUNT(*) AS total_transactions
FROM payments
GROUP BY payment_type
ORDER BY average_order_value DESC;
```

```sql
SELECT payment_type, ROUND(AVG(payment_value), 2) AS average_order_value, COUNT(*) AS total_transactions FROM payments GROUP BY payment_type ORDER BY average_order_value DESC;
```

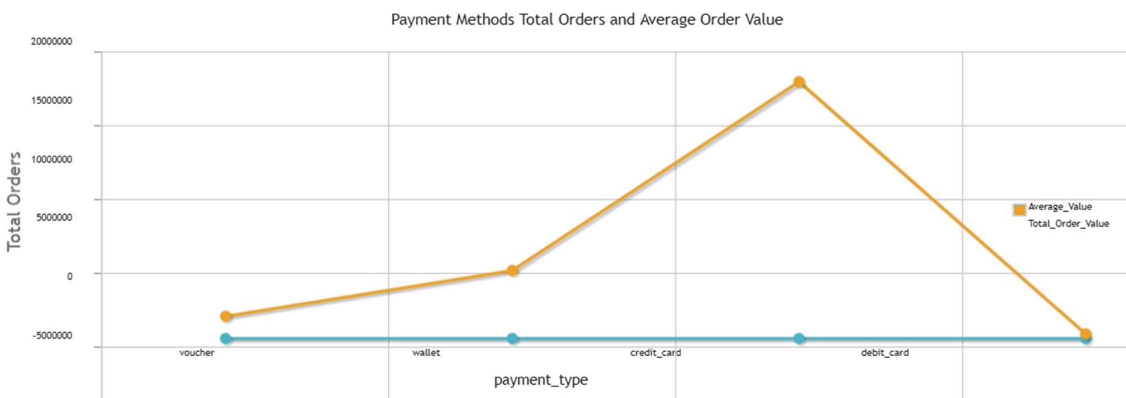
payment_type	average_order_value	total_transactions
voucher	310.42	4886
wallet	268.48	17227
credit_card	266.34	65511
debit_card	246.46	1283



```sql
SELECT
    payment_type,
    AVG(payment_value) AS Average_Value,
    SUM(payment_value) AS Total_Order_Value
FROM payments
GROUP BY payment_type
ORDER BY Average_Value DESC
```

```sql
SELECT payment_type, AVG(payment_value) AS Average_Value, SUM(payment_value) AS Total_Order_Value FROM payments GROUP BY payment_type ORDER BY Average_Value DESC;
```

payment_type	Average_Value 1	Total_Order_Value
voucher	310.417382	1516699.33
wallet	268.479922	4625103.62
credit_card	266.341891	17448323.62
debit_card	246.462027	316210.78



Conta il totale degli ordini

```sql
SELECT COUNT(*) FROM orders;
```

COUNT(*)
89316

Conta il totale degli ordini annullati

```sql
SELECT COUNT(order_status) AS total_canceled_orders
FROM orders
WHERE order_status = 'canceled';
```

```sql
SELECT COUNT(order_status) AS total_canceled_orders FROM orders WHERE order_status = 'canceled';
```

total_canceled_orders
409

Annulla gli ordini con lo stato 'canceled'

```
DELETE FROM orders
WHERE order_status = 'canceled'
```

Conta il totale degli ordini dopo l'annullamento di quelli con lo stato 'canceled

```
SELECT COUNT(*) FROM orders;
```

COUNT(*)
88907